

Technology Stack

Date	25 June 2026
Team ID	
Project Name	Rising Waters – A Flood Prediction
Maximum Marks	2 Marks

Technology Stack Details

S.NO	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Builds a responsive and user-friendly interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side	Python, Flask	Processes user requests, loads the trained ML model, performs flood prediction, and returns results to the web application.

3	Database / Data Storage	CSV Dataset, Joblib (.pkl) Model Files	Stores the training dataset and saves the trained ML model for prediction without retraining
4	Local Development / Execution	Anaconda Navigator, Jupyter Notebook, VS Code	Used for data preprocessing, model training, testing, debugging, and running the Flask application locally.
5	Version Control & CI/CD	Git, GitHub	Maintains source code versions, enables collaboration, and provides project backup.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib	Used for data preprocessing, visualization, machine learning model development, evaluation, and model serialization.